Solution

Matrix Theory, Quiz 5A — March 20, 2026

Given the matrix

$$A = \begin{bmatrix} 2 & -1 & 0 & 2 \\ 0 & 0 & 2 & -1 \\ -2 & 1 & -2 & -1 \end{bmatrix},$$

answer the following.

(1) Find a basis for the column space of A.
(2) Find a basis for the row space of A.
(3) Find a basis for the null space of A.
(4) Extend the basis given in (1) to a basis for $\mathbb{R}^3$.
(5) Determine whether the vector $\mathbf{v} = \begin{bmatrix} 2 \\ -1 \\ -1 \end{bmatrix}$ is in the column space of A. If yes, write $\mathbf{v}$ as a linear combination of the basis given in (1).

3 reduce A to RREF

$$\to \begin{bmatrix} 2 & -1 & 0 & 2 \\ 0 & 0 & 2 & -1 \\ 0 & 0 & -2 & 1 \end{bmatrix} \to \begin{bmatrix} 2 & -1 & 0 & 2 \\ 0 & 0 & 1 & -1/2 \\ 0 & 0 & 0 & 0 \end{bmatrix} \to \begin{bmatrix} 1 & -1/2 & 0 & 1 \\ 0 & 0 & 1 & -1/2 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

this is RREF

c_1 c_3

1 (1) $\left\{ \begin{bmatrix} 2 \\ 0 \\ -2 \end{bmatrix}, \begin{bmatrix} 0 \\ 2 \\ -2 \end{bmatrix} \right\}$ basis for ColSp A

1 (2) $\{[1, -1/2, 0, 1], [0, 0, 1, -1/2]\}$ ← basis Row Sp A

1 (3) Null sp $= b \begin{bmatrix} 1/2 \\ 1 \\ 0 \\ 0 \end{bmatrix} + d \begin{bmatrix} -1 \\ 0 \\ 1/2 \\ 1 \end{bmatrix}$

basis Null Sp A $= \left\{ \begin{pmatrix} 1/2 \\ 1 \\ 0 \\ 0 \end{pmatrix}, \begin{pmatrix} -1 \\ 0 \\ 1/2 \\ 1 \end{pmatrix} \right\}$

2 (4) $\begin{bmatrix} 2 & 0 & 1 \\ 0 & 2 & 0 \\ -2 & -2 & 0 \end{bmatrix} \to \begin{bmatrix} 2 & 0 & 1 \\ 0 & 2 & 0 \\ 0 & -2 & 1 \end{bmatrix}$

$\to \begin{bmatrix} 2 & 0 & 1 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$ - basis for $\mathbb{R}^3$

$\left\{ \begin{bmatrix} 2 \\ 0 \\ -2 \end{bmatrix}, \begin{bmatrix} 0 \\ 2 \\ -2 \end{bmatrix}, \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} \right\}$

2 Yes: Coord. vect $= \begin{bmatrix} 1 \\ -1/2 \end{bmatrix}$

(5) $\begin{bmatrix} 2 \\ 0 \\ -2 \end{bmatrix} - 1/2 \begin{bmatrix} 0 \\ 2 \\ -2 \end{bmatrix} = \begin{bmatrix} 2 \\ -1 \\ -1 \end{bmatrix}$

Solution

Matrix Theory, Quiz 5B — March 20, 2026

Given the matrix

$$B = \begin{bmatrix} 2 & -1 & 0 & -2 \\ -2 & 1 & -2 & 1 \\ 0 & 0 & 2 & 1 \end{bmatrix},$$

answer the following.

(1) Find a basis for the column space of B.
(2) Find a basis for the row space of B.
(3) Find a basis for the null space of B.
(4) Extend the basis given in (1) to a basis for $\mathbb{R}^3$.
(5) Determine whether the vector $\mathbf{v} = \begin{bmatrix} -2 \\ 1 \\ 1 \end{bmatrix}$ is in the column space of B. If yes, write $\mathbf{v}$ as a linear combination of the basis given in (1).

3 reduce B to RREF

$$B \to \begin{bmatrix} 2 & -1 & 0 & -2 \\ 0 & 0 & -2 & -1 \\ 0 & 0 & 2 & 1 \end{bmatrix} \to \begin{bmatrix} 2 & -1 & 0 & -2 \\ 0 & 0 & 1 & 1/2 \\ 0 & 0 & 0 & 0 \end{bmatrix} \to \begin{bmatrix} 1 & -1/2 & 0 & -1 \\ 0 & 0 & 1 & 1/2 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$ ← this is RREF

(1) basis col sp $= \{c_1, c_3\} = \left\{ \begin{pmatrix} 2 \\ -2 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ -2 \\ 2 \end{pmatrix} \right\}$

(2) R_1, R_2 of RREF = basis Row sp $= \{ [1, -1/2, 0, -1], [0, 0, 1, 1/2] \}$

(3) Null sp $= b \begin{pmatrix} 1/2 \\ 1 \\ 0 \\ 0 \end{pmatrix} + d \begin{pmatrix} 1 \\ 0 \\ -1/2 \\ 1 \end{pmatrix}$ basis Null sp $= \left\{ \begin{pmatrix} 1/2 \\ 1 \\ 0 \\ 0 \end{pmatrix}, \begin{pmatrix} 1 \\ 0 \\ -1/2 \\ 1 \end{pmatrix} \right\}$

(4) Start with $[c_1, c_3, e_1] \xrightarrow{\text{reduce}} \begin{bmatrix} 2 & 0 & 1 \\ -2 & -2 & 0 \\ 0 & 2 & 0 \end{bmatrix} \to \begin{bmatrix} 2 & 0 & 1 \\ 0 & -2 & 1 \\ 0 & 2 & 0 \end{bmatrix} \xrightarrow[R_2+R_3]{R_3} \begin{bmatrix} 2 & 0 & 1 \\ 0 & 2 & 0 \\ 0 & 0 & 1 \end{bmatrix}$

Rank = 3

this is basis for $\mathbb{R}^3$: $\left\{ \begin{pmatrix} 2 \\ -2 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ -2 \\ 2 \end{pmatrix}, \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} \right\}$

(5) Yes Solve $[c_1\ c_3 \mid c_4] = \left[\begin{array}{cc|c} 2 & 0 & -2 \\ -2 & -2 & 1 \\ 0 & 2 & 1 \end{array}\right] \to \left[\begin{array}{cc|c} 1 & 0 & -1 \\ 0 & -2 & -1 \\ 0 & 2 & 1 \end{array}\right]$

2

$\to \left[\begin{array}{cc|c} 1 & 0 & -1 \\ 0 & 1 & +1/2 \\ 0 & 0 & 0 \end{array}\right]$ coord vector $= \begin{bmatrix} -1 \\ 1/2 \end{bmatrix}$

$$-\begin{bmatrix} 2 \\ -2 \\ 0 \end{bmatrix} + 1/2 \begin{bmatrix} 0 \\ -2 \\ 2 \end{bmatrix} = \begin{bmatrix} -2 \\ 1 \\ 1 \end{bmatrix}$$